

# 2012 European guideline for the management of anogenital warts

## Abstract

### Background

Although new HPV vaccines have been developed and are in the process of implementation, anogenital warts remain a very frequent problem in clinical practice.

### Objective

To update previously published European guidelines for the management of anogenital warts.

### Methods

A systematic review of randomized controlled trials for anogenital warts was conducted. The primary data were analyzed and collated, and the findings were formulated into a clinical guideline. The draft was reviewed by the IUSTI Europe Editorial Board and made available online for public comment; feedback was incorporated into the final version.

### Results

The data confirm that only surgical therapies achieve primary clearance rates approaching 100%. Recurrences—包括 previously treated sites or new lesions—occur after all treatments, with recurrence rates often 20–30% or higher. All treatment modalities are associated with local skin reactions such as itching, burning, erosions, and pain.

### Conclusions

Physicians managing patients with genital warts should develop individualized treatment algorithms based on local practices and guidelines. Patient management should include clinical review at least every 4 weeks, with a change in therapy if there is an inadequate response. Patients with first-episode

warts should be offered screening for other sexually transmitted infections. Management must also include partner notification and health promotion.

## Aetiology and transmission

Anogenital warts (also known as genital warts, condylomata acuminata, or condylomas) are benign proliferative lesions caused primarily by human papillomavirus (HPV) types 6 and 11, which are detected in >90% of cases. Co-infection with high-risk HPV types, such as HPV 16, is common. Genital warts are transmitted sexually, with an estimated transmission rate of approximately 60% between untreated sexual partners.

This guideline was developed on behalf of: - The European Branch of the International Union against Sexually Transmitted Infections (IUSTI Europe) - The European Dermatology Federation (EDF) - The Union of European Medical Specialists (UEMS)

## Clinical features

### *Symptoms*

Most patients are asymptomatic aside from the visible presence of lesions. When symptoms occur, they may include itching, bleeding, fissuring, or dyspareunia.

### *Physical signs*

Lesions typically appear at sites subject to trauma during intercourse. They may be solitary, but multiple lesions (usually 5–15) measuring 1–5 mm in diameter are more common. Warts can coalesce into larger plaques, particularly in immunosuppressed individuals or those with diabetes.

In uncircumcised men, the preputial cavity—including the glans penis, coronal sulcus, frenulum, and inner foreskin—is most frequently affected. In circumcised

men, the penile shaft is more commonly involved. Other sites include the urethral meatus, pubis, scrotum, groin, perineum, perianal area, and anal canal.

In women, warts may occur on the fourchette, labia minora, labia majora, pubis, clitoris, urethral meatus, perineum, perianal region, anal canal, introitus, vagina, and ectocervix.

Intra-anal warts are most frequently seen in individuals who practice receptive anal intercourse, though they can occur without it. Warts are rarely found proximal to the dentate line.

Anogenital warts vary in color from pink or salmon red to white, greyish-white, or various shades of brown (pigmented lesions). Pigmentation is more commonly observed on the labia majora, penile shaft, pubis, groin, perineum, and perianal area.

## Complications

### *Physical and psychosexual implications*

Anogenital warts are often perceived as disfiguring and can significantly affect sexual well-being. Patients may experience anxiety, guilt, anger, and reduced self-esteem. Common concerns include potential effects on fertility and cancer risk.

### *Pre-cancer and cancer*

Premalignant conditions—such as vulval intraepithelial neoplasia (VIN), anal intraepithelial neoplasia (AIN), and penile intraepithelial neoplasia (PIN)—as well as invasive cancers (vulval, anal, and penile) may coexist with anogenital warts, develop within them, or be initially misdiagnosed as warts. Clinicians should maintain a high index of suspicion for such conditions, especially in persistent, atypical, or treatment-resistant lesions.